Hypothesis

- Hypothesis
  - Eg: Imagine a teacher who wants to check if giving students a practice in Kahoot improves their test scores.
  - **sample** of 10 students, give them practice in Kahoot , and compare their scores to the **average score** from the academic records

Hypothesis

- Null hypothesis (H_0): $\mu1=\mu2$ (no difference in population means)
- Alternative (Ha) : $\mu1\neq\mu2$ (means differ)
- Eg:
  - Null Hypothesis (H_0): Kahoot has no effect. Mean score = class average (say, 70).
  - Alternative Hypothesis (Ha): Kahoot changes the mean score (mean ≠ 70).

Some important points

- Compare **p-value** with **α**:
  - If **p-value < α** → **Reject H_0** (evidence supports H_1).
  - If **p-value ≥ α** → **Fail to reject H_0** (not enough evidence against H_0).

Critical Value Approach

- **Right-tailed test:**
  - If statistic > critical value → Reject H_0. Else → Retain H_0.
- **Left-tailed test:**
  - If statistic < critical value → Reject H_0. Else → Retain H_0.

HYPOTHESIS TEST FOR POPULATION MEAN UNDER UNKNOWN POPULATION VARIANCE: *t*-TEST

- When we test a population mean, the test statistic depends on the distribution of the sample mean.
- If population variance is known, we use the z-test.
- If population variance is unknown, we use the t-test, because we estimate the variance from the sample.

•Suppose the population is normally distributed.
•If the variance is unknown, we estimate it using the sample standard deviation SSS.
•The standardized statistic then follows a t-distribution with (n−1) degrees of freedom.

t-test

$$t = \frac{\bar{X} - \mu}{S / \sqrt{n}}$$

Where:

- $\bar{X}$ = sample mean
- μ = hypothesized population mean (under H_0)
- S = sample standard deviation
- n = sample size
- Degrees of freedom = $n - 1$

The t-test is used when the population follows a normal distribution and the population standard deviation σ is unknown and is estimated from the sample. t-test is a robust test for violation of normality of the data as long as the data is close to symmetry and there are no outliers.

EXAMPLE 6.5

Aravind Productions (AP) is a newly formed movie production house based out of Mumbai, India. AP was interested in understanding the production cost required for producing a Bollywood movie. The industry believes that the production house will require at least INR 500 million (50 crore) on average. It is assumed that the Bollywood movie production cost follows a normal distribution. Production cost of 40 Bollywood movies in millions of rupees are shown in Table 6.7. Conduct an appropriate hypothesis test at $a = 0.05$ to check whether the belief about average production cost is correct.

TABLE 6.7 Production cost of Bollywood movies

601	627	330	364	562	353	583	254	528	470
125	60	101	110	60	252	281	227	484	402
408	601	593	729	402	530	708	599	439	762
292	636	444	286	636	667	252	335	457	632

- **Solution:** It is given that the production cost of Bollywood movies follows a normal distribution; however, the standard deviation of the population is not known and we need to estimate the standard deviation value from the sample. Thus, we have to use the *t*-test for testing the hypothesis. From the sample data in Table 6.7 we get the following values:

$$n = 40,\ \bar{X} = 429.55, \text{ and } S = 195.0337$$

The null and alternative hypotheses are

$$H_0: \mu \leq 500$$

$$H_A: \mu > 500$$

The corresponding test statistic is

$$t\text{-statistic} = \frac{\bar{X} - \mu}{S/\sqrt{n}} = \frac{429.55 - 500}{195.0337/\sqrt{40}} = -2.2845$$

- Note that this is a one-tailed test (right-tailed) and the critical *t*-value at $\alpha = 0.05$ under right-tailed test, *t*critical = 1.6848 [in Excel TINV(2α, *df*) will return right-tailed critical value at significance of α, in this example $\alpha = 0.05$, the corresponding critical *t*-value using Excel function is TINV(0.1, 39) = 1.6848, that is the critical value is 1.6848].
- Since *t*-statistic value is less than the critical *t*-value, we retain the null hypothesis. The *t*-statistic value and critical value for the *t*-test are shown in Figure 6.10.

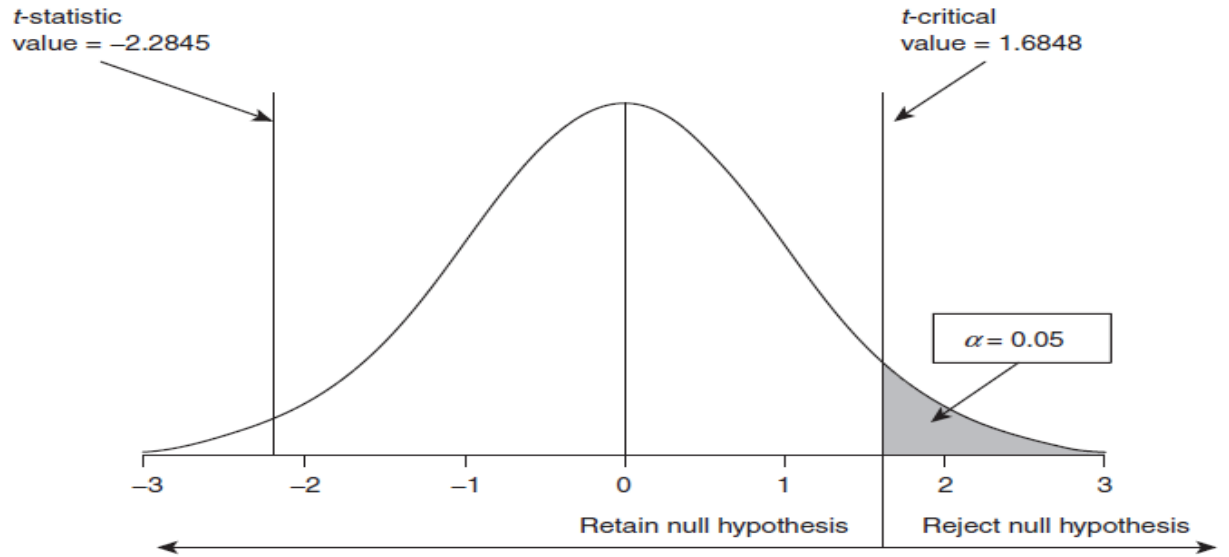


FIGURE 6.10 Critical value, *t*-statistic value for *t*-test in Example 6.5.

df	Critical Values				
1-tailed	0.1	0.05	0.025	0.01	0.005
2-tailed	0.2	0.1	0.05	0.02	0.01
1	3.0777	6.3138	12.7062	31.8205	63.6567
2	1.8856	2.9200	4.3027	6.9646	9.9248
3	1.6377	2.3534	3.1824	4.5407	5.8409
4	1.5332	2.1318	2.7764	3.7469	4.6041
5	1.4759	2.0150	2.5706	3.3649	4.0321
6	1.4398	1.9432	2.4469	3.1427	3.7074
7	1.4149	1.8946	2.3646	2.9980	3.4995
8	1.3968	1.8595	2.3060	2.8965	3.3554
9	1.3830	1.8331	2.2622	2.8214	3.2498
10	1.3722	1.8125	2.2281	2.7638	3.1693
11	1.3634	1.7959	2.2010	2.7181	3.1058
12	1.3562	1.7823	2.1788	2.6810	3.0545
13	1.3502	1.7709	2.1604	2.6503	3.0123
14	1.3450	1.7613	2.1448	2.6245	2.9768
15	1.3406	1.7531	2.1314	2.6025	2.9467
16	1.3368	1.7459	2.1199	2.5835	2.9208
17	1.3334	1.7396	2.1098	2.5669	2.8982
18	1.3304	1.7341	2.1009	2.5524	2.8784
19	1.3277	1.7291	2.0930	2.5395	2.8609
20	1.3253	1.7247	2.0860	2.5280	2.8453
21	1.3232	1.7207	2.0796	2.5176	2.8314
22	1.3212	1.7171	2.0739	2.5083	2.8188
23	1.3195	1.7139	2.0687	2.4999	2.8073
24	1.3178	1.7109	2.0639	2.4922	2.7969
25	1.3163	1.7081	2.0595	2.4851	2.7874
26	1.3150	1.7056	2.0555	2.4786	2.7787
27	1.3137	1.7033	2.0518	2.4727	2.7707
28	1.3125	1.7011	2.0484	2.4671	2.7633

df	Critical Values				
1-tailed	0.1	0.05	0.025	0.01	0.005
2-tailed	0.2	0.1	0.05	0.02	0.01
32	1.3086	1.6939	2.0369	2.4487	2.7385
33	1.3077	1.6924	2.0345	2.4448	2.7333
34	1.3070	1.6909	2.0322	2.4411	2.7284
35	1.3062	1.6896	2.0301	2.4377	2.7238
36	1.3055	1.6883	2.0281	2.4345	2.7195
37	1.3049	1.6871	2.0262	2.4314	2.7154
38	1.3042	1.6860	2.0244	2.4286	2.7116
39	1.3036	1.6849	2.0227	2.4258	2.7079
40	1.3031	1.6839	2.0211	2.4233	2.7045
41	1.3025	1.6829	2.0195	2.4208	2.7012
42	1.3020	1.6820	2.0181	2.4185	2.6981
43	1.3016	1.6811	2.0167	2.4163	2.6951
44	1.3011	1.6802	2.0154	2.4141	2.6923
45	1.3006	1.6794	2.0141	2.4121	2.6896
46	1.3002	1.6787	2.0129	2.4102	2.6870
47	1.2998	1.6779	2.0117	2.4083	2.6846
48	1.2994	1.6772	2.0106	2.4066	2.6822
49	1.2991	1.6766	2.0096	2.4049	2.6800
50	1.2987	1.6759	2.0086	2.4033	2.6778
60	1.2958	1.6706	2.0003	2.3901	2.6603
70	1.2938	1.6669	1.9944	2.3808	2.6479
80	1.2922	1.6641	1.9901	2.3739	2.6387
90	1.2910	1.6620	1.9867	2.3685	2.6316
100	1.2901	1.6602	1.9840	2.3642	2.6259
110	1.2893	1.6588	1.9818	2.3607	2.6213
120	1.2886	1.6577	1.9799	2.3578	2.6174
130	1.2881	1.6567	1.9784	2.3554	2.6142
140	1.2876	1.6558	1.9771	2.3533	2.6114

EXAMPLE 6.6

According to statistics released by the Department of Civil Aviation, the average delay of flights is equal to 16.8 minutes, flight delays are assumed to follow a normal distribution. However, from a sample of 50 flights, the average delay was estimated to be 19.5 minutes and the sample standard deviation was 6.6 minutes. Conduct a hypothesis test to disprove the claim that the average delay is equal to 16.8 minutes at $a = 0.01$.

Solution:

Given $n = 50,\ \bar{X} = 19.5, S = 6.6$

Null and alternative hypotheses are

$$H_0: \mu = 16.8$$

$$H_A: \mu \neq 16.8$$

The corresponding *t*-statistic value is

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n}} = \frac{19.5 - 16.8}{6.6/\sqrt{50}} = 2.8927$$

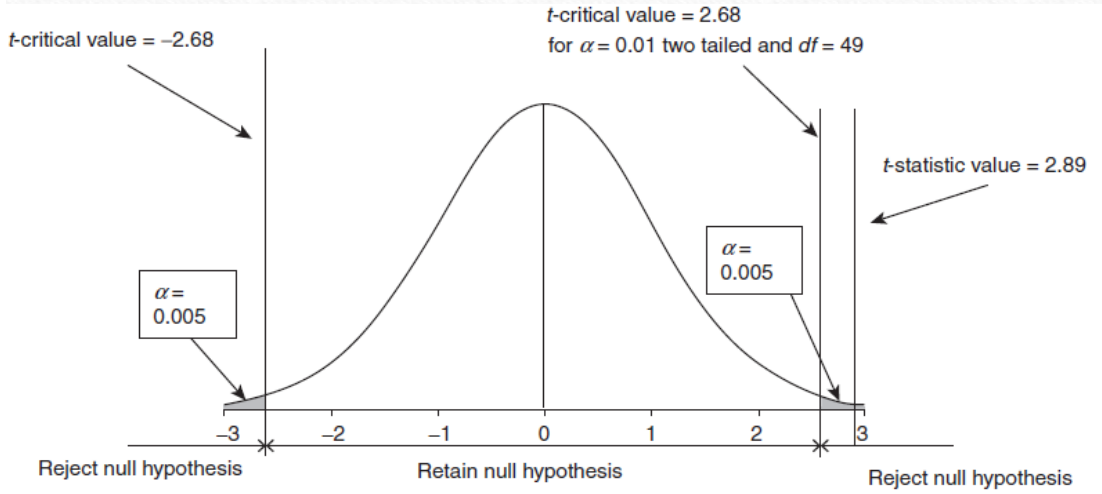


FIGURE 6.11 *t*-statistic, *t*-critical, rejection and acceptance regions for Example 6.6.

- The critical *t*-value for two-tailed *t*-test when *a* = 0.01 and degrees of freedom = 49 is 2.67 [in Excel, TINV(0.01, 49) = 2.68 or T.INV.2T(0.01, 49) = 2.68].
- Since the calculated *t*-statistic value is greater than the *t*-critical value, we reject the null hypothesis.
- The corresponding *p*-value is 0.0057 [in excel T.DIST(*t*-statistic value, degrees of freedom, tails) returns the *p*-value, T.DIST.2T(2.8927, 49) = 0.0057]. The values of *t*-statistic, *t*-critical value, rejection and retention regions are shown in Figure 6.11.

df	Critical Values				
1-tailed	0.1	0.05	0.025	0.01	0.005
2-tailed	0.2	0.1	0.05	0.02	0.01
1	3.0777	6.3138	12.7062	31.8205	63.6567
2	1.8856	2.9200	4.3027	6.9646	9.9248
3	1.6377	2.3534	3.1824	4.5407	5.8409
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10	1.3722	1.8125	2.2281	2.7638	3.1693
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13	1.3502	1.7709	2.1604	2.6503	3.0123
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15	1.3406	1.7531	2.1314	2.6025	2.9467
16	1.3368	1.7459	2.1199	2.5835	2.9208
17	1.3334	1.7396	2.1098	2.5669	2.8982
18	1.3304	1.7341	2.1009	2.5524	2.8784
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23	1.3195	1.7139	2.0687	2.4999	2.8073
24	1.3178	1.7109	2.0639	2.4922	2.7969
25	1.3163	1.7081	2.0595	2.4851	2.7874
26	1.3150	1.7056	2.0555	2.4786	2.7787
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28	1.3125	1.7011	2.0484	2.4671	2.7633

df	Critical Values				
1-tailed	0.1	0.05	0.025	0.01	0.005
2-tailed	0.2	0.1	0.05	0.02	0.01
32	1.3086	1.6939	2.0369	2.4487	2.7385
33	1.3077	1.6924	2.0345	2.4448	2.7333
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35	1.3062	1.6896	2.0301	2.4377	2.7238
36	1.3055	1.6883	2.0281	2.4345	2.7195
37	1.3049	1.6871	2.0262	2.4314	2.7154
38	1.3042	1.6860	2.0244	2.4286	2.7116
39	1.3036	1.6849	2.0227	2.4258	2.7079
40	1.3031	1.6839	2.0211	2.4233	2.7045
41	1.3025	1.6829	2.0195	2.4208	2.7012
42	1.3020	1.6820	2.0181	2.4185	2.6981
43	1.3016	1.6811	2.0167	2.4163	2.6951
44	1.3011	1.6802	2.0154	2.4141	2.6923
45	1.3006	1.6794	2.0141	2.4121	2.6896
46	1.3002	1.6787	2.0129	2.4102	2.6870
47	1.2998	1.6779	2.0117	2.4083	2.6846
48	1.2994	1.6772	2.0106	2.4066	2.6822
49	1.2991	1.6766	2.0096	2.4049	2.6800
50	1.2987	1.6759	2.0086	2.4033	2.6778
60	1.2958	1.6706	2.0003	2.3901	2.6603
70	1.2938	1.6669	1.9944	2.3808	2.6479
80	1.2922	1.6641	1.9901	2.3739	2.6387
90	1.2910	1.6620	1.9867	2.3685	2.6316
100	1.2901	1.6602	1.9840	2.3642	2.6259
110	1.2893	1.6588	1.9818	2.3607	2.6213
120	1.2886	1.6577	1.9799	2.3578	2.6174
130	1.2881	1.6567	1.9784	2.3554	2.6142
140	1.2876	1.6558	1.9771	2.3533	2.6114

HYPOTHESIS TESTING FOR POPULATION MEAN WITH KNOWN VARIANCE: *Z*-TEST

One-Sample Z-Test

Z-Test

- A **Z-test** is used when we want to test a **claim (hypothesis)** about a **population parameter** (like population mean or proportion).
- It is applied when the **population variance (σ^2) is known**.

One-Sample Z-Test

- When the test is carried out using **only one sample**, it is called the **One-Sample Z-Test**.
- It checks whether the **sample mean** is significantly different from the **population mean**.

- **Central Limit Theorem (CLT)**
- For a **large sample size (n > 30)** taken from an independent and identically distributed (i.i.d.) population:
  - The **sampling distribution of the sample mean** is approximately **normal**.
  - Mean of sampling distribution = μ (population mean)
  - Standard deviation of sampling distribution = $\sigma/\sqrt{n}$
- **Standardized Value (Z)**
- $Z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$
- Where:
- $\bar{X}$ = sample mean
- μ = population mean (hypothesized)
- σ = population standard deviation
- n = sample size

TABLE 6.4 Critical value for different values of α

	Approximate Critical Values		
α	Left-Tailed Test	Right-Tailed Test	Two-Tailed Test
0.1	−1.28	1.28	−1.64 and 1.64
0.05	−1.64	1.64	−1.96 and 1.96
0.01	−2.33	2.33	−2.58 and 2.58

Excel Functions for Critical Values

- **Left-tailed test:** `=NORMSINV(α)`
- **Right-tailed test:** `=NORMSINV(1 – α)`
- **Two-tailed test:**
  - Left critical = `=NORMSINV(α/2)`
  - Right critical = `=NORMSINV(1 – α/2)`

TABLE 6.5 Condition for rejection of null hypothesis H_0

Type of Test	Condition	Decision
Left-tailed test	Z-statistic < Critical value	Reject H_0
	Z-statistic ≥ Critical value	Retain H_0
Right-tailed test	Z-statistic > Critical value	Reject H_0
	Z-statistic ≤ Critical value	Retain H_0
Two-tailed test	$\lvert Z\text{-statistic}\rvert > \lvert\text{Critical Value}\rvert$	Reject H_0
	$\lvert Z\text{-statistic}\rvert \leq \lvert\text{Critical Value}\rvert$	Retain H_0

One-sample Z-test is used when

1. Testing the value of population mean when population standard deviation is known.

2. The population is a normal distribution and the population variance is known.

3. The sample size is large and the population variance is known. That is, the assumption of

normal distribution can be relaxed for large samples ($n > 30$).

Example 6.1

An agency based out of Bangalore claimed that the average monthly disposable income of families living in Bangalore is greater than INR 4200 with a standard devi ation of INR 3200. From a random sample of 40,000 families, the average disposable income was estimated as INR 4250. Assume that the population standard deviation is INR 3200. Conduct an appropriate hypothesis test at 95% confidence level (a = 0.05) to check the validity of the claim by the agency

IMPORTANT

In contexts such as this, we set alternative hypothesis as the statement that we would like to prove.

Claim: Average disposable income is more than INR 4200

Let μ and σ denote the mean and standard deviation in the population. The corresponding null and alternative hypotheses are

$$H_0: \mu \leq 4200$$
$$H_A: \mu > 4200$$

Since we know the population standard deviation, we can use the Z-test. The corresponding Z-statistic is given by

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} = \frac{4250 - 4200}{3200 / \sqrt{40000}} = 3.125$$

- This is a right-tailed test. The corresponding *Z*-critical value at $a = 0.05$ for right-tailed test is approximately 1.64 [in Excel NOMRSINV(1 − *a*) that is NORMSINV(0.95) gives the critical value for the right-tailed test].
- Since the calculated *Z*-statistic value is greater than the *Z*-critical value, we reject the null hypothesis. The corresponding *p*-value = 0.00088 [*p*-value in Excel is given by 1 − NORMSDIST(*Z*-statistic value), that is 1 – NORMSDIST(3.125) in this case].
- The critical value, *Z*-statistic value, and the corresponding *p*-value are shown in Figure 6.5.

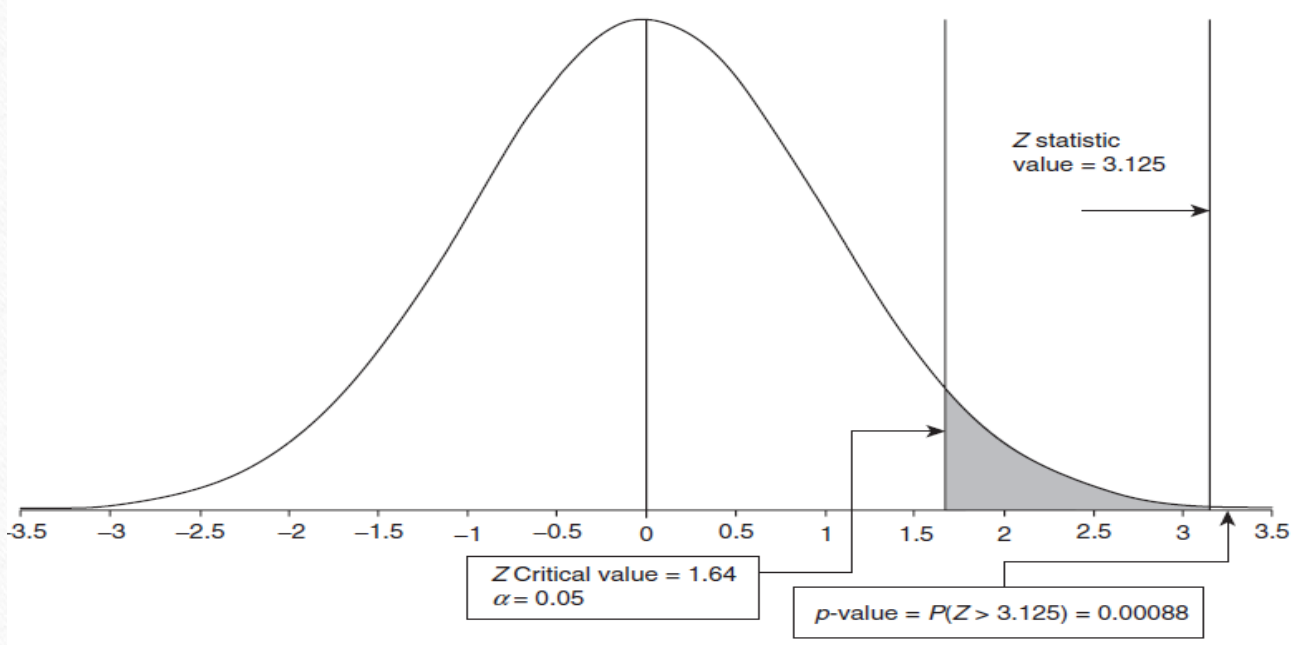


FIGURE 6.5 Critical value, *Z*-statistic value, and corresponding *p*-value.

Z-statistic measures the standardized difference between estimated value of mean and the hypothesis value of mean. Z = 3.125 implies that the sample mean is at 3.125 standard deviations away from the hypothesized population mean given that the null hypothesis is true

Example 6.2

A passport office claims that the passport applications are processed within 30 days of submitting the application form and all necessary documents. Table 6.6 shows processing time of 40 passport applicants. The population standard deviation of the processing time is 12.5 days. Conduct a hypothesis test at significance level $a = 0.05$ to verify the claim made by the passport office.

TABLE 6.6 Passport processing time

16	16	30	37	25	22	19	35	27	32
34	28	24	35	24	21	32	29	24	35
28	29	18	31	28	33	32	24	25	22
21	27	41	23	23	16	24	38	26	28

Solution

Null and alternative hypotheses in this case are given by

$$H_0: \mu \geq 30$$
$$H_A: \mu < 30$$

From the data in Table 6.6, the estimated sample mean is 27.05 days.

The standard deviation of the sampling distribution $\sigma / \sqrt{n} = 12.5 / \sqrt{40} = 1.9764$.

The value of Z-statistic is given by

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} = \frac{27.05 - 30}{12.5 / \sqrt{40}} = -1.4926$$

- The critical value of left-tailed test for $\alpha = 0.05$ is -1.644. Since the critical value is less than the *Z*-statistic value, we fail to reject the null hypothesis.
- The *p*-value for $Z = -1.4926$ is (NORMSDIST(*Z*-statistic value)) 0.06777 which is greater than the value of α. That is, there is no strong evidence against null hypothesis so we retain the null hypothesis, which is $\mu \geq 30$.
- Figure 6.6 shows the calculated *Z*-statistic value and the rejection region.

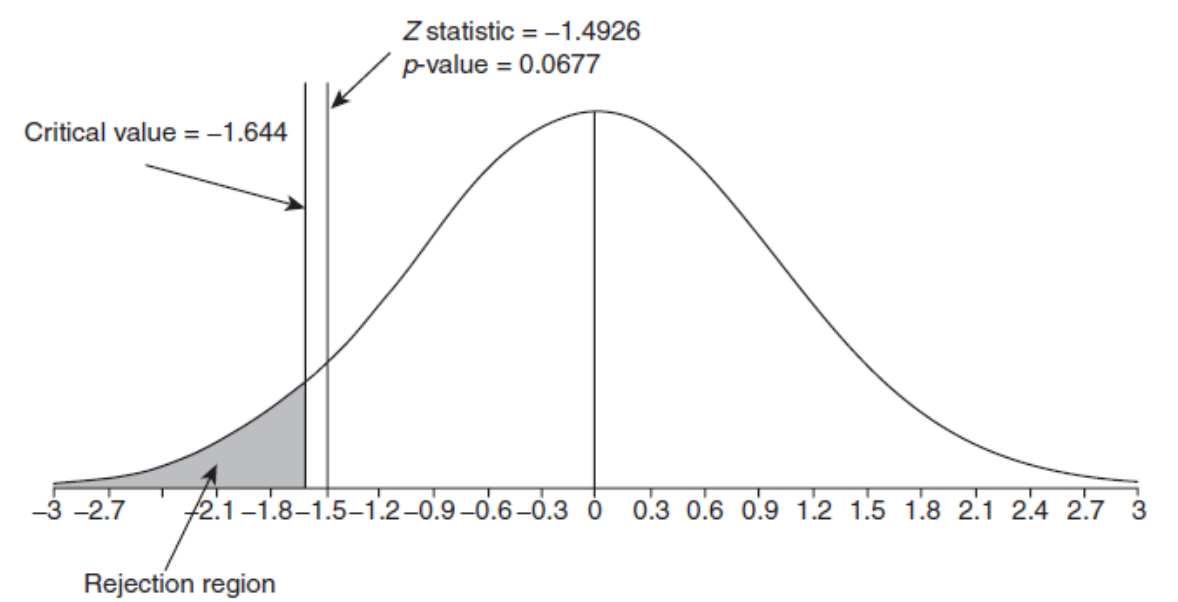


FIGURE 6.6 Left-tailed test for Example 6.2.

According to the company IQ Research, the average Intelligence Quotient (IQ) of Indians is 82 derived based on a research carried out by Professor Richard Lynn, a British Professor of Psychology, using data collected from 2002 to 2006 (Source: IQ Research1). The population standard deviation of IQ is estimated as 11.03. Based on a sample of 100 people from India, the sample IQ was estimated as 84. Conduct an appropriate hypothesis test at $\alpha = 0.05$ to validate the claim of IQ Research (that average IQ of Indians is 82)

- (a) Hypothesis test: It is given that m = 82, s = 11.03, n = 100, and The null and alternative hypotheses in this case are

$$H_0: \mu = 82$$
$$H_A: \mu \neq 82$$

- Since the direction of alternative hypothesis is both ways, we have a two-tailed t-test. The test statistics is given by

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} = \frac{84 - 82}{11.03 / \sqrt{100}} = 1.8132$$

For a two-tailed test, the critical values at a/2 = 0.025 are −1.96 and 1.96 [in Excel NORMSINV(0.025) = −1.96 and NORMSINV(1 − 0.025) = 1.96]. Since the calculated Z-statistic value is within the critical values, we fail to reject the null hypothesis (retain the null hypothesis). Figure 6.7 shows the rejection regions and the Z-statistic value in this case. Since the Z-statistic value is 1.8132 and falls on the right tail, we first calculate normal distribution beyond 1.8132 which is equal to 0.0348. Since this is a two-tailed test, the p-value is twice the area to the right side of the Z-statistic value, which is = 0.0698, that is the p-value in this case is 0.0698.

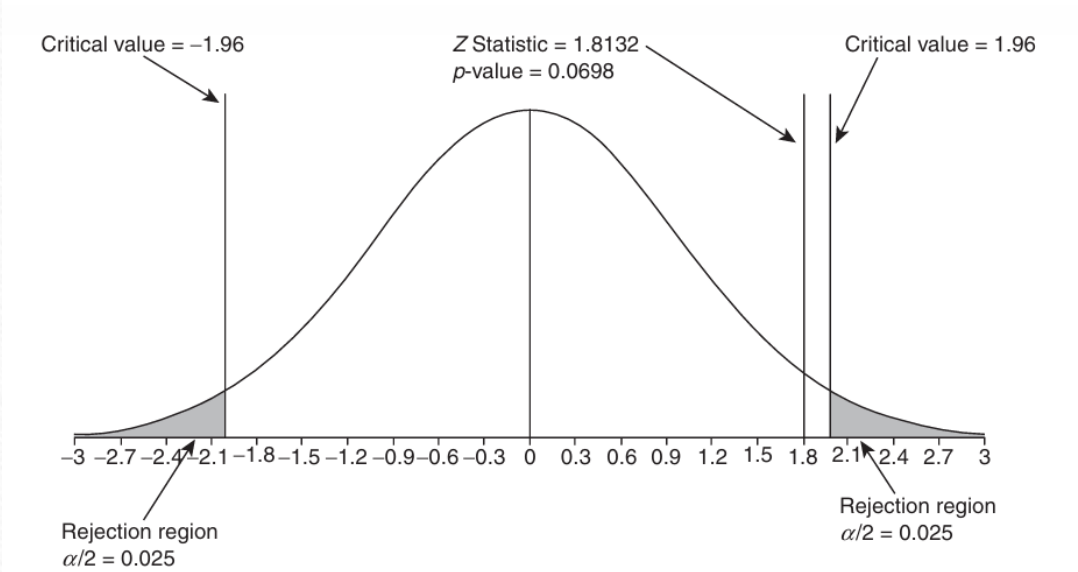


FIGURE 6.7 *Z*-statistic, critical values, and the rejection region for Example 6.3.

Procedure for testing hypothesis using t-Test 2 sample

Step 1: Form Hypothesis (Null and alternate)

Step 2: Select a distribution

- The size of sample is small
- Standard deviation or variance is not known
- 2 different samples with different size, select independent t-Test
- Assuming population to be normal

Step 3: Calculate mean and variance

Step 4: Check if the variance is equal or unequal.

Procedure for testing hypothesis using t-test

Step 5: Calculate t-Statistics :T_{cal}

Step 6: Find T_{tab} value of t-statistics (Refer T-distribution table)

- Degree of freedom
- Significance level
- One tail/two tailed

Step 7: Compare T_{cal} and T_{tab}

- If $T_{cal} > T_{tab}$, reject Null hypothesis

Paired Sample t-test

- In a paired t-test, the data related to the parameter is captured twice from the same subject, once before the intervention and once after intervention.
- Alternatively, the paired t-test can be used for comparing two different interventions such as two different promotion strategies applied on the same subject (price discount versus bundling of items).

Examples of paired t-test

1. Body weight of subjects before and after attending a yoga training program.
2. Cholesterol levels of subjects before and after attending meditation training.
3. Amount of time spent by subjects on the internet before and after marriage.
4. Quantity of alcohol consumed by people before and after breakup.
5. Level of cortisol among students during and after exam.

Paired t-test formula

- Note that, in the above examples, we are observing a population parameter value on the same subject before and after intervention. Assume that the mean difference in the estimated parameter value before and after the treatment is D, and the corresponding standard deviation of difference is Sd . Let μ_d be the hypothesized mean difference. Then the statistic defined in Eq. below follows a t-distribution with (n − 1) degrees of freedom.

$$\frac{D-\mu_d}{S_d/\sqrt{n}}$$

- Here we assume that the differences follow a normal distribution

Example: Table 6.8 shows data on alcohol consumption before and after breakup. Conduct a paired t-test to check whether the alcohol consumption is more after the breakup (that is μd > 0) at 95% confidence (α = 0.05).

Solution: The mean difference, that is mean of (X2 – X1), is 11.5 and the corresponding sample standard deviation is 95.67.

The null and alternative hypotheses are (when the claim is that the difference is greater than zero):

$$H_o: \mu_d \leq 0$$
$$H_A: \mu_d > 0$$

The value of test statistic is

$$t = \frac{D - \mu_d}{S_d / \sqrt{n}} = \frac{11.5 - 0}{95.6757 / \sqrt{20}} = 0.5375$$

Sr. No	Before Breakup (X1)	After Breakup (X2)	Difference (X2-X1)
1	470	408	-62
2	354	439	85
3	496	321	-175
4	351	437	86
5	349	335	-14
6	449	344	-105
7	378	318	-60
8	359	492	133
9	469	531	62
10	329	417	88
11	389	358	-31
12	497	391	-106
13	493	398	-95
14	268	394	126
15	445	508	63
16	287	399	112
17	338	345	7
18	271	341	70
19	412	326	-86
20	335	467	132

- The critical t-value for one-tailed test when $\alpha = 0.05$ and $df = 19$ is 1.7291 [in Excel T.INV(0.05, 19) = 1.7291]. Since the t-statistic value is 0.5375, which is less than the critical value, we retain the null hypothesis and conclude the difference in alcohol consumption is not greater than 0 before and after breakup. The corresponding p-value is 0.70

df	Critical Values				
1-tailed	0.1	0.05	0.025	0.01	0.005
2-tailed	0.2	0.1	0.05	0.02	0.01
1	3.0777	6.3138	12.7062	31.8205	63.6567
2	1.8856	2.9200	4.3027	6.9646	9.9248
3	1.6377	2.3534	3.1824	4.5407	5.8409
4	1.5332	2.1318	2.7764	3.7469	4.6041
5	1.4759	2.0150	2.5706	3.3649	4.0321
6	1.4398	1.9432	2.4469	3.1427	3.7074
7	1.4149	1.8946	2.3646	2.9980	3.4995
8	1.3968	1.8595	2.3060	2.8965	3.3554
9	1.3830	1.8331	2.2622	2.8214	3.2498
10	1.3722	1.8125	2.2281	2.7638	3.1693
11	1.3634	1.7959	2.2010	2.7181	3.1058
12	1.3562	1.7823	2.1788	2.6810	3.0545
13	1.3502	1.7709	2.1604	2.6503	3.0123
14	1.3450	1.7613	2.1448	2.6245	2.9768
15	1.3406	1.7531	2.1314	2.6025	2.9467
16	1.3368	1.7459	2.1199	2.5835	2.9208
17	1.3334	1.7396	2.1098	2.5669	2.8982
18	1.3304	1.7341	2.1009	2.5524	2.8784
19	1.3277	1.7291	2.0930	2.5395	2.8609
20	1.3253	1.7247	2.0860	2.5280	2.8453
21	1.3232	1.7207	2.0796	2.5176	2.8314
22	1.3212	1.7171	2.0739	2.5083	2.8188
23	1.3195	1.7139	2.0687	2.4999	2.8073
24	1.3178	1.7109	2.0639	2.4922	2.7969
25	1.3163	1.7081	2.0595	2.4851	2.7874
26	1.3150	1.7056	2.0555	2.4786	2.7787
27	1.3137	1.7033	2.0518	2.4727	2.7707
28	1.3125	1.7011	2.0484	2.4671	2.7633

df	Critical Values				
1-tailed	0.1	0.05	0.025	0.01	0.005
2-tailed	0.2	0.1	0.05	0.02	0.01
32	1.3086	1.6939	2.0369	2.4487	2.7385
33	1.3077	1.6924	2.0345	2.4448	2.7333
34	1.3070	1.6909	2.0322	2.4411	2.7284
35	1.3062	1.6896	2.0301	2.4377	2.7238
36	1.3055	1.6883	2.0281	2.4345	2.7195
37	1.3049	1.6871	2.0262	2.4314	2.7154
38	1.3042	1.6860	2.0244	2.4286	2.7116
39	1.3036	1.6849	2.0227	2.4258	2.7079
40	1.3031	1.6839	2.0211	2.4233	2.7045
41	1.3025	1.6829	2.0195	2.4208	2.7012
42	1.3020	1.6820	2.0181	2.4185	2.6981
43	1.3016	1.6811	2.0167	2.4163	2.6951
44	1.3011	1.6802	2.0154	2.4141	2.6923
45	1.3006	1.6794	2.0141	2.4121	2.6896
46	1.3002	1.6787	2.0129	2.4102	2.6870
47	1.2998	1.6779	2.0117	2.4083	2.6846
48	1.2994	1.6772	2.0106	2.4066	2.6822
49	1.2991	1.6766	2.0096	2.4049	2.6800
50	1.2987	1.6759	2.0086	2.4033	2.6778
60	1.2958	1.6706	2.0003	2.3901	2.6603
70	1.2938	1.6669	1.9944	2.3808	2.6479
80	1.2922	1.6641	1.9901	2.3739	2.6387
90	1.2910	1.6620	1.9867	2.3685	2.6316
100	1.2901	1.6602	1.9840	2.3642	2.6259
110	1.2893	1.6588	1.9818	2.3607	2.6213
120	1.2886	1.6577	1.9799	2.3578	2.6174
130	1.2881	1.6567	1.9784	2.3554	2.6142
140	1.2876	1.6558	1.9771	2.3533	2.6114

A researcher believes that people drink more coffee on Mondays than other days of the week. Based on a sample of 50 coffee drinkers, the mean difference was estimated as 14 ml and the corresponding standard deviation was 8.5 ml. Conduct an appropriate hypothesis test at $\alpha = 0.1$ to check the claim that people drink on average 10 ml more coffee on Mondays compared to other days of the week

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- We are given n = 50, D = 14, Sd = 8.5, and μd = 10. The null and alternative hypotheses are

$$H_0: \mu_d \leq 10$$

$$H_A: \mu_d > 10$$

- The test statistic is

$$t = \frac{D - \mu_d}{S_d / \sqrt{n}} = \frac{14 - 10}{8.5 / \sqrt{50}} = 3.3275$$

- Critical value of t for a = 0.1 and df = 49 is 1.2990 [in Excel T.INV(1 − 0.1, 49) = 1.2990]. Since t-statistic value is greater than t-critical value, we reject the null hypothesis. That is, there is evidence from the data that people drink at least 10 ml more coffee on Mondays than other days. The corresponding p-value is 0.000834 [in Excel, T.DIST.RT(3.3275, 49) = 0.000834]

Comparing two sample test Z test and t Test

COMPARING TWO POPULATIONS: TWO-SAMPLE Z- AND *t*-TEST

1. Difference in Two Population Means when Population Standard Deviations are Known: Two-Sample *Z*-Test
2. Difference in Two Population Means when Population Standard Deviations are Unknown and Believed to be Equal: Two-Sample t-Test
3. Difference in Two Population Means when Population Standard Deviations are Unknown and Not Equal: Two-Sample *t*-Test with Unequal Variance

Difference in Two Population Means when Population Standard Deviations are Known: Two-Sample *Z*-Test

In this case we make the following assumptions:

1. The sample sizes (say n_1 and n_2) of two samples drawn from two populations are large (say at least 30) and the corresponding standard deviations σ_1 and σ_2 are known.

2. The samples are drawn from two normally distributed populations with corresponding standard deviations σ_1 and σ_2 known.

Difference in Two Population Means when Population Standard Deviations are Known: Two-Sample Z-Test

Assume that μ_1 and μ_2 are the population means. Our interest is to check a hypothesis on difference between μ_1 and μ_2, that is $(\mu_1 - \mu_2)$. If $\bar{X}_1$ and $\bar{X}_2$ are the estimated mean values from two samples drawn from two populations, the statistic $(\bar{X}_1 - \bar{X}_2)$ follows a standard normal distribution with mean $(\mu_1 - \mu_2)$ and standard deviation $\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$, where n_1 and n_2 are the sample sizes of two samples. The corresponding Z-statistic is given by

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \tag{6.10}$$

EXAMPLE 6.9: The Dean of St Peter School of Management Education (SPSME) believes that the graduating students with specialization in Marketing earn at least INR 5000 more per month than the students with specialization in Operations Management. To verify his belief, the Dean collected a sample data from his graduating students, given in Table 6.9. Conduct an appropriate hypothesis test at $\alpha = 0.05$ to check whether the difference in monthly salary is at least 5000 more for students with marketing specialization compared to operations specialization. Assume that the salary of students with marketing specialization and operations specialization follow normal distribution.

TABLE 6.9 Sample values on marketing and operations students

Specialization	Sample Size	Estimated Mean Salary (in Rupees) per Month	Population Standard Deviation
Marketing	120	67,500.00	7,200
Operations	45	58,950.00	4,600

Solution:

We have $n_1 = 120$, $n_2 = 45$, $\overline{X}_1 = 67,500$, $\overline{X}_2 = 58,950$, $\sigma_1 = 7,200$ and $\sigma_2 = 4,600$. The null and alternative hypotheses are

$$H_0: \mu_1 - \mu_2 \leq 5000$$

$$H_A: \mu_1 - \mu_2 > 5000$$

The corresponding test statistic value is

$$Z = \frac{(67500 - 58950) - 5000}{\sqrt{\frac{7200^2}{120} + \frac{4600^2}{45}}} = \frac{3550}{949.85} = 3.7374$$

The critical value of Z at $\alpha = 0.05$ is 1.64 [= NORMSINV(1 − 0.05)]. Since the Z-statistic value is higher than the Z-critical value, we reject the null hypothesis. The corresponding p-value is 9.29×10^{-05}.

Difference in Two Population Means when Population Standard Deviations are Unknown and Believed to be Equal: Two-Sample t-Test

In this section we discuss the hypothesis test for difference in two population means when the standard deviations of the populations are unknown. Hence we need to estimate them from the samples drawn from these two populations. An additional assumption we make here is that the standard deviation of two populations are equal (however, unknown). Then the sampling distribution of the difference in estimated means $(\bar{X}_1 - \bar{X}_2)$ follows a t-distribution with $(n_1 + n_2 - 2)$ degrees of freedom with mean $(\mu_1 - \mu_2)$ and standard deviation

Equal-Variance (Pooled) t-Test

$$\sqrt{S_p^2\left(\frac{1}{n_1}+\frac{1}{n_2}\right)} \tag{6.11}$$

where S_p^2 is the pooled variance of two samples and is given by

$$S_p^2 = \frac{(n_1-1)S_1^2+(n_2-1)S_2^2}{(n_1+n_2-2)} \tag{6.12}$$

The corresponding t-statistic is

$$t = \frac{(\bar{X}_1-\bar{X}_2)-(\mu_1-\mu_2)}{\sqrt{S_p^2\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}} \tag{6.13}$$

Equal-Variance (Pooled) t-Test – General formula assuming $H_0 : \mu_1 = \mu_2.$

- For comparing 2 groups
- Test whether two independent groups have significantly different means.

$$T = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}, \quad s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

- $\bar{x}_1, \bar{x}_2$: sample means
- s_1^2, s_2^2: sample variances
- n_1, n_2: sample sizes
- s_p: pooled standard deviation

- Degree of Freedom : df=n1+n2−2

Unequal-Variance t-Test (Welch t Test)

The formula is diff in Ref Book 3 with some changes but the answer obtained from below and the Ref Book are same. For easy understanding we will use the below formula

- Variance across two groups are not equal

$$T = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \quad , \quad \mathrm{df} = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{1}{n_1 - 1}\left(\frac{s_1^2}{n_1}\right)^2 + \frac{1}{n_2 - 1}\left(\frac{s_2^2}{n_2}\right)^2}$$

- The t-values will be
  - +ve : if mean of group 1 is larger than the mean of group 2
  - -ve : if mean of group 2 is larger than mean of group 1

- When looking at sample summaries, statisticians calculate the ratio between the two groups. The standard critical values used to switch from equal to unequal variance are:
- **Standard Deviation Ratio of 2.0:** The standard deviation of one group should not be more than **2 times** larger than the standard deviation of the other group.

$$\frac{\text{Larger } s}{\text{Smaller } s} < 2 \Longrightarrow \textbf{Equal Variance Assumed (Pooled)}$$

- **Variance Ratio of 4.0:** Because variance is just standard deviation squared s^2, this translates to a variance ratio threshold of **4**.

$$\frac{\text{Larger } s^2}{\text{Smaller } s^2} < 4 \Longrightarrow \textbf{Equal Variance Assumed (Pooled)}$$

A company makes a claim that children (in the age group between 7 and 12) who drink their health drink will grow taller than the children who do not drink that health drink. Data in Table 6.10 shows average increase in height over one-year period from two groups: one drinking the health drink and the other not drinking the health drink. At $\alpha = 0.05$, test whether the increase in height for the children who drink the health drink is at least 1.2 cm.

TABLE 6.10 Data related in increase in heights from different groups

Group	Sample Size	Increase in Height (in cm) during the Test Period	Standard Deviation Estimated from Sample
Drink health drink	80	7.6 cm	1.1 cm
Do not drink health drink	80	6.3 cm	1.3 cm

Solution

We have $n_1 = 80$, $n_2 = 80$, $\bar{X}_1 = 7.6$, $\bar{X}_2 = 6.3$, $\sigma_1 = 1.1$, and $\sigma_2 = 1.3$. (Note that the sample standard deviations need not be same for samples although the population standard deviations are same.)

The null and alternative hypotheses are

$$H_0: \mu_1 - \mu_2 \leq 1.2$$

$$H_A: \mu_1 - \mu_2 > 1.2$$

Solution Contd.

Pooled variance is

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{(n_1 + n_2 - 2)} = \frac{79 \times 1.1^2 + 79 \times 1.3^2}{80 + 80 - 2} = 1.45$$

The t-statistic is

$$t = \frac{(7.6 - 6.3) - 1.2}{\sqrt{1.45\left(\frac{1}{80} + \frac{1}{80}\right)}} = 0.5252$$

The t-critical value for one-tailed t-test when $\alpha = 0.05$ and degrees of freedom = 158 (80 + 80 – 2) is 1.6546. Since the calculated t-statistic value is less than t-critical value we retain the null hypothesis. That is, the difference between two groups is less than 1.2 and the corresponding right-tailed test has a p-value of 0.3.

Example

- A tech company wants to see if a new AI-assisted training program speeds up the time it takes for new hires to push their first code to production compared to their traditional onboarding pipeline.
  The company tracks two independent groups of new hires over six months. Because fewer people went through the experimental AI track, the sample sizes are uneven, and the spread of completion times varies significantly.
  At $\alpha = 0.05$, test the claim that the AI-assisted training program significantly reduces the average days to first production push compared to the traditional program.

Collected Sample Data:

- **Group 1 (Traditional Pipeline):**
  - Sample Size (n_1): **45** employees
  - Average Days to Push ($\bar{X}_1$): **28.4** days
  - Sample Standard Deviation (s_1): **3.1** days
- **Group 2 (AI-Assisted Track):**
  - Sample Size (n_2): **18** employees
  - Average Days to Push ($\bar{X}_2$): **21.2** days
  - Sample Standard Deviation (s_2): **7.8** days

Why this is a Welch's Test (Checking the Thresholds)

1. **Standard Deviation Ratio:** The larger standard deviation ($s_2 = 7.8$) divided by the smaller standard deviation ($s_1 = 3.1$) is $\frac{7.8}{3.1} = 2.52$. Since this is **greater than 2.0**, it violates the equal variance rule of thumb.
2. **Variance Ratio:** Squaring them gives variances of 60.84 versus 9.61. The ratio is $\frac{60.84}{9.61} = 6.33$. Since this is **greater than 4.0**, equal variance cannot be assumed.
3. **Sample Sizes:** The sample sizes are heavily unbalanced (45 vs 18), making a pooled equal-variance t-test highly unreliable.

- **Null Hypothesis :**The AI program does not speed up the process

$$\mu_1 \leq \mu_2$$

- **Alternative Hypothesis :**The AI program significantly reduces days to first push)

$$\mu_1 > \mu_2$$

- **Based on threshold using formula:**

$$T = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \quad , \quad \mathrm{df} = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{1}{n_1 - 1}\left(\frac{s_1^2}{n_1}\right)^2 + \frac{1}{n_2 - 1}\left(\frac{s_2^2}{n_2}\right)^2}$$

Example Contd.

- Calculate the intermediate variance terms:

$$\frac{s_1^2}{n_1} = \frac{3.1^2}{45} = \frac{9.61}{45} \approx 0.213556$$

$$\frac{s_2^2}{n_2} = \frac{7.8^2}{18} = \frac{60.84}{18} = 3.380000$$

- Compute the t-statistic:

$$T = \frac{28.4 - 21.2}{\sqrt{0.213556 + 3.380000}} = \frac{7.2}{\sqrt{3.593556}} = \frac{7.2}{1.895668} \approx \mathbf{3.798}$$

Example Contd.

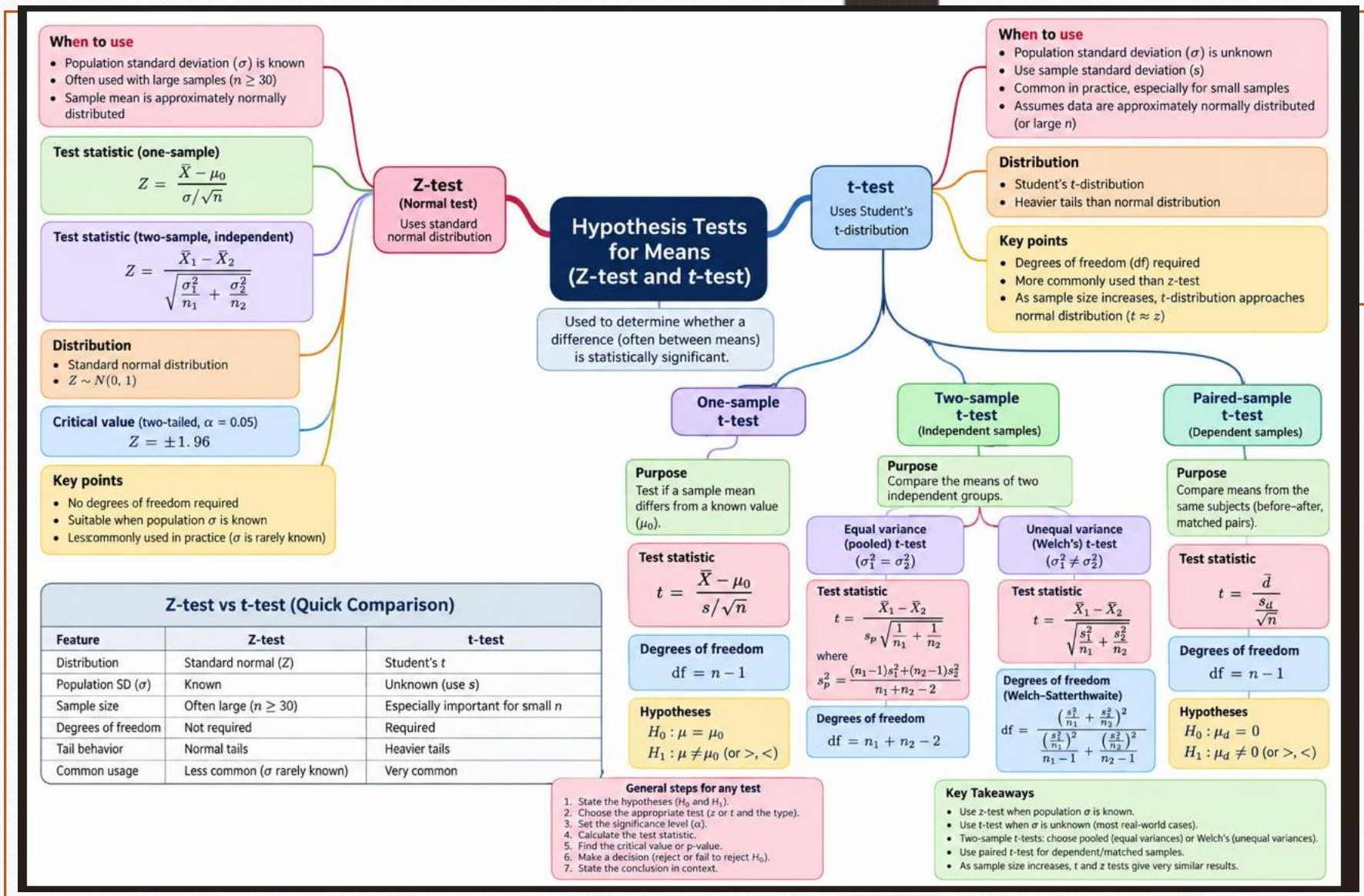
- **Compute the Degrees of Freedom df-**
- **Rounding off df to 19**

$$df = \frac{(0.213556 + 3.380000)^2}{\frac{1}{45-1}(0.213556)^2 + \frac{1}{18-1}(3.380000)^2}$$

$$df = \frac{(3.593556)^2}{\frac{1}{44}(0.045606) + \frac{1}{17}(11.424400)}$$

$$df = \frac{12.913645}{0.001037 + 0.672024} = \frac{12.913645}{0.673061} \approx 19.19$$

- Find the critical value from a student's t-distribution table at is 1.729
- Since t-statistical > t-critical reject null hypothesis.
- Conclude that the AI-assisted training program significantly reduces the average days to the first production push.



This slide is just to summarize the different t and z test scenarios. Kindly verify the formulas.

For Hypothesis chapter we are referring reference book 3 chapter 6 excluding portions 6.10, 6.11, 6.12.